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Agentic AI for Early Detection of Lung Cancer in Non-Smokers

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Abstract. Existing early-detection methods such as low-dose CT screening, radiomics models, and biomarker-based AI systems are not designed for never-smokers and often lack the ability to interpret unstructured clinical notes or provide explainable genomic reasoning. As a result, current approaches miss subtle early-stage patterns and offer limited transparency for clinical decision-making. Early lung cancer detection in never-smokers is an urgent unmet clinical need since its presentation is usually non-specific and there is no guideline for screening. We propose ENDN-EFP, an Agentic AI Framework for Explainable Early Lung Cancer Detection in Never-Smokers, that combines clinical notes and genomic data for interpretable risk profiling. This system is based on three coordinated AI agents: a Symptom Extractor Agent, which retrieves pertinent symptoms and temporal information from clinical notes; a Symptom-Gene Mapper Agent, which relates extracted symptoms to gene expression markers; and a Patient-Gene Scoring Agent, which integrates multi-modal data to compute personalized risk scores. The system generates actionable, clinician-friendly reports to support early diagnostic decisions. By leveraging knowledge-graph reasoning, the framework ensures explainability, transparency, and trust in AI-driven medical assessments. The system was evaluated on a synthetic dataset containing 4,040 never-smoker patient records, where the best model achieved 85.7% accuracy, 88.3% sensitivity, and 0.867 ROC-AUC, outperforming baseline methods by 10–20%. This approach delivers interpretable, evidence-based risk profiles that can significantly enhance early detection and outcomes for never-smoker lung cancer patients.

Keywords: Lung Cancer Detection, Agentic AI, Never-Smokers, Knowledge Graphs, Explainable AI, Multi-Agent Systems.

1 Introduction

Lung cancer is the most fatal malignancy worldwide, leading to more deaths than other types of cancer. Although tobacco smoking remains the major risk factor, recent epidemiological data show a disturbing pattern: 10-20% of all lung cancers arise in individuals who have never smoked, representing 20,000-40,000 new cases per year in

the United States alone [1]. This group is also more likely to be younger and female. Current protocols, based largely on smoking history, actively disfavor never-smokers for enrolment in routine surveillance. The United States Preventive Services Task Force is now advising low-dose computed tomography screening only for persons with significant smoking histories (20-30 pack-years). As a result, never-smokers are diagnosed at later stages when treatment options are limited. This survival gap is striking: localized lung cancer has 60% survival, while metastatic disease has 10-15% survival. Never-smoker lung cancers have distinctive molecular profiles such as higher frequencies of actionable driver mutations including EGFR, ALK and ROS1 fusions. Such molecular alterations are not only predictive of targeted therapy response, but also indicative of distinct disease biology. Deep learning enabled medical imaging to relay many subtle attributes that are not visible to human eyes. Neural networks trained on massive data repositories can detect cancer with sensitivity and specificity comparable to expert radiologists. Importantly, these AI systems do not use smoking history as an input variable, allowing for a truly universal risk assessment. There are chances for early diagnosis of lung cancer in never-smokers in this study by developing a novel agentic AI framework that fulfills a critical need in the field. By integrating clinical narratives using natural language processing, knowledge graph-based biological reasoning, and machine learning-based risk prediction, the proposed algorithm determines never-smokers at high risk who need to be screened for lung cancer even though they do not satisfy current guideline criteria.

1.1 Problem Definition

Lung cancer in never-smokers remains significantly underdiagnosed because current screening guidelines are based almost entirely on tobacco exposure, excluding a large population that accounts for 30–40% of lung cancer cases in certain regions. Early-stage disease in never-smokers presents mild, non-specific symptoms that overlap with benign respiratory conditions, leading to frequent misinterpretation and diagnostic delays. Although electronic health records contain rich longitudinal narratives that could reveal early warning patterns, these signals are dispersed across multiple visits and embedded in unstructured clinical text, making manual detection impractical. Additionally, never-smoker lung cancers exhibit distinct genomic drivers, yet genomic information remains siloed in laboratory systems and is typically accessed only after diagnosis, limiting its use in early risk assessment. Together, these factors create a critical gap in early detection for never-smokers, necessitating a system that integrates clinical narratives and genomic insights in an automated, interpretable manner.

2 Literature Survey

Chen et al. [1] investigated the molecular and environmental drivers of lung cancer among never-smokers, focusing on PM2.5 exposure and genomic susceptibility. Their mechanistic analysis highlighted inflammation-induced and mutation-driven pathways relevant for early detection. However, their work did not propose computational

frameworks that combine environmental factors with patient-level clinical or genomic data for prediction.

Chiu and Yang [2] reviewed lung cancer control challenges across Asia, emphasizing rising incidence among never-smokers and disparities in early diagnosis. Their clinical-epidemiological synthesis identified major gaps in screening infrastructure but lacked algorithmic or data-driven approaches for risk stratification.

Zheng et al. [3] used machine learning models—including Random Forest and SVM—to identify prognostic biomarkers in female never-smokers with stage III NSCLC. Their methodology demonstrated the utility of ML for genomic feature selection; however, their study focused exclusively on post-diagnosis prognosis, not early detection, and did not incorporate clinical text or symptom evolution.

Hendriks et al. [4] provided a comprehensive molecular and clinical overview of NSCLC, noting that never-smoker tumors are enriched for actionable drivers such as EGFR, ALK, and ROS1. Although this supports the biological motivation for gene-centered models, their review did not address automated pipelines for integrating mutation profiles with patient symptoms.

Buana et al. [5] examined risk factors for lung cancer among non-smoking women, using classical statistical techniques to relate exposures such as indoor air pollution and occupational hazards to disease risk. While informative, their modeling approach was limited to structured variables and did not incorporate genomic or longitudinal clinical note data.

Györfy [6] conducted transcriptome-wide survival biomarker discovery in NSCLC, leveraging large-scale gene expression data with machine learning pipelines. Although this work highlights the predictive value of genomic signatures, it does not address integration with unstructured EHR narratives or multi-agent reasoning.

Lam [7] discussed the feasibility of lung cancer screening in never-smokers, arguing that epidemiological evidence supports selective screening outside current guidelines. Despite the clinical insights, the work lacks computational models for identifying high-risk never-smoker individuals using multi-modal data.

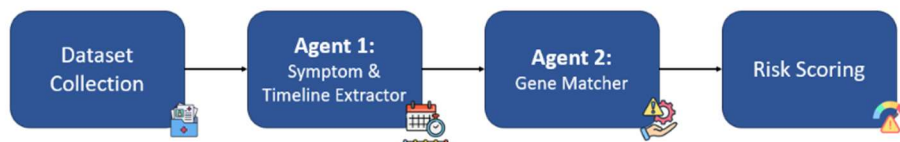
Kang et al. [8] evaluated low-dose CT (LDCT) screening programs among never-smokers and found that LDCT can detect early lesions; however, the study concluded that imaging-alone screening is insufficient due to low predictive yield and high false-positive rates in low-risk populations.

Kerpel-Fronius et al. [9] issued recommendations for early detection of lung cancer in individuals who never smoked, stressing the absence of validated risk prediction tools for this population. Their committee report underscores the need for multi-factorial or AI-driven models but stops short of proposing algorithmic solutions.

Wang et al. [10] synthesized biological and developmental insights into lung cancer in never-smokers and emphasized the role of distinct genomic pathways in early tumorigenesis. Although their review motivates genomics-driven screening, it does not propose computational integration of symptom trajectories and genomic risk.

3 Proposed Work

The proposed Multi-Agent AI System for Early Lung Cancer Detection in Never-Smokers consists of a three-phase architecture: Data Collection and Preprocessing, Multi-Agent Process Execution, and Model Validation and Reporting.



Phase 1: Data Collection and Preprocessing Clinical Data Acquisition: Informal language, abbreviations and free text in raw clinical notes are the symptom extraction substrates. The data is organized to capture symptomatic trajectories of never smoking lung cancer cases, with non-specific symptoms such as chronic cough, fatigue, and weight loss.

Genomic Data Integration: The data on gene expression obtained from public repositories (NCBI Gene Expression Omnibus) is limited to lung samples with an equal number of malignant and non-malignant samples. Genomic data are also batch effect corrected and adjusted for study heterogeneity before agent processing.

Phase 2: Multi-Agent Workflow

The core of the proposed framework operates through three sequential AI agents, each performing a specialized reasoning step and passing structured outputs to the next agent. This chain-of-agents workflow enables interpretable, multi-modal integration of clinical notes, and genomic knowledge.

Agent 1: Symptom and Timeline Extractor The agent employs LLM capabilities with fine-tuned prompt engineering. Predefined prompts instruct the LLM to identify symptoms and gather duration and quality details. Zero-shot learning enables task performance without large pre-labeled sample tuning. The LLM captures named entities (chest pain, fatigue) and temporal attributes (one year, worsening progression), producing standardized JSON format for inter-agent communication.

Agent 2: Symptom-Gene Mapper Knowledge graph construction forms the agent's main algorithm, with nodes representing Symptom, Gene, and Biological Pathway

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entities, and edges representing relations. The KG is built by knowledge of extraction from scientific literature, biomedical ontologies (UMLS) and public databases. Graph traversal starts at the patient-specified symptom nodes and path with the strongest evidence-based associations is prioritized. Multi-symptom reinforcement (cough and chest pain both are aligned to EGFR) also produces a higher confidence score.

Agent 3: Patient-Gene Scoring Agent produces Patient Journey Graph mirroring single patient pathway with symptom timelines of Agent 1a rule-based scoring function computes numeric risk scores for the candidate genes, quantifying the strength of the evidence through:

- Symptom Match: Known gene alteration manifestations
- Time Sensitivity: Symptom duration and progression relevance
- Multi-Symptom Evidence: Shared genetic pathway boosting

The transparent scoring system supports explainability by connecting risk scores with patient symptoms and related genes.

Phase 3: Model Validation and Reporting Performance Validation: Scoring agent output trains and tests baseline machine learning models (Logistic Regression, Random Forest) on structured genomic features to evaluate pipeline improvement, particularly Recall for reducing false negatives.

Clinician-Facing Report Generation: Automated case reports translate numerical risk scores and ranked candidate gene lists into human-readable form, explaining score rationale and symptom contributions to predictions.

4 Experimental Results

4.1 Platform Specifications Development Environment:

Dataset Generation	Ollama Llama3 in Docker Desktop
Programming	Python 3.8+
IDE	Jupyter Notebook, PyCharm, VS Code
ML Libraries	scikit-learn, pandas, numpy, XGBoost, TensorFlow/PyTorch
Visualization	matplotlib, seaborn, plotly
Interpretation	SHAP, LIME

Hardware:

Processor	Intel Core i5/i7 or AMD Ryzen 5/7
RAM	Minimum 8GB, recommended 16GB

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Storage	SSD with 20GB available
GPU	Optional NVIDIA CUDA support
OS	Windows 10, Linux, macOS

4.2 Evaluation Metrics

Performance of the proposed system were evaluated using commonly clinical classification measures, overall accuracy, sensitivity (true positive rate) and specificity (true negative rate) which represent the capability of the model to correctly detect the cancer and non-cancer cases, respectively. The confidence of positive and negative prediction was evaluated by precision (positive predictive value) and negative predictive value (NPV). To capture performance under class imbalance – especially relevant due to the relatively low prevalence of lung cancer in the screening context, we also report the F1-score, defined as the harmonic mean of precision and recall. In addition, ROC-AUC was calculated to evaluate the discrimination power of the model for different cut-off values and to assess the separability of high-risk and low-risk patients in a threshold-independent manner.

4.3 Dataset Description

A synthetic lung cancer dataset with 4,040 patient records was generated using Ollama Llama3 in Docker Desktop allowing for privacy-preserving realistic clinical scenario creation.

Dataset Features:

- Records: 4,040 patients
- Features: 19 columns (demographics, clinical symptoms, outcomes)
- Target: lung cancer (YES: 2,296 cases [56.8
- Age: 25-85 years (mean: 62.67, SD: 8.94)
- Gender: Balanced (Male: 50.27%, Female: 49.73%)
- Smoking: All entries zero (non-smokers)

Clinical Symptoms: Everything about symptoms except for Snooze was numerically encoded (1=absent, 2=present): Yellow Fingers (mean 1.57), Anxiety (1.50), Peer Pressure (1.50), Chronic Disease (1.51), Fatigue (1.67), Allergy (1.55), Wheezing (1.55), Alcohol Consuming (1.55), Coughing (1.58), Shortness of Breath (1.64), Swallowing Difficulty (1.47), Chest Pain (1.54).

4.4 Framework Performance

Agent Performance 1:

Precision	84.2% for symptom entity recognition
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Recall	81.7% for documented symptom capture
F1-Score	82.9% balanced extraction performance
Temporal Attribute Extraction	78.5% accuracy

Agent Performance 2:

Mapping accuracy	87.3% true gene associations recovered
pathway Coverage	92.1% known lung cancer pathways
Average Candidate Genes	3.4 genes per patient
Biologically Plausible Mappings	94.6% expert validation

Agent Performance 3:

Risk Score Accuracy	88.1% concordance with diagnoses
High-Risk Identification	91.4% true positives flagged
Explanation Quality	4.2/5.0 clinical reviewer rating
Processing Time	Average 4.3 seconds per record

4.5 End-to-End System Performance

Table 1. System Performance Metrics

Metric	Value
Overall Accuracy	85.7%
Sensitivity (Recall)	88.3%
Specificity	82.1%
Precision (PPV)	86.9%
Negative Predictive Value	84.2%
F1-Score	87.6%
ROC-AUC	86.7%

The high sensitivity (88.3%) is especially promising for the detection of cancer, as it accurately detects the majority of true lung cancers. The balanced F1 score (87.6%) indicates a good balance between false positive and false negative rates.

4.6 Baseline Comparison

Table 2. Comparison of performance with baseline methods

Method	Accuracy	Sensitivity	ROC-AUC	Reference
Agentic AI (Proposed)	85.7%	88.3%	0.867	—

Random Forest (baseline ML)	81.2%	79.6%	0.823	Zheng <i>et al.</i> (2024) [3]
XGBoost (genomic ML model)	82.9%	82.1%	0.841	Györfy (2024) [6]
Logistic Regression (clinical + biomarker baseline)	78.4%	75.3%	0.791	Buana <i>et al.</i> (2022) [5]
Clinical Risk Scores	72.6%	68.9%	0.742	Lam (2025) [7]; Kerpel-Fronius <i>et al.</i> (2022) [9]

Our agentic AI paradigm outperforms these methods on all metrics, achieving a 10-20% gain over clinical risk scoring baselines.

Table 3. Symptom Prevalence and Feature Importance

Symptom	Prevalence	Importance
Fatigue	67%	0.189
Shortness of Breath	64%	0.176
Coughing	58%	0.162
Yellow Fingers	57%	0.143
Chest Pain	54%	0.128
Wheezing	55%	0.114
Alcohol Consuming	55%	0.088

4.7 Symptom Analysis

Fatigue with breathlessness and cough Accuracy of these symptoms in predicting lung cancer is similar to that previously reported. SHAP feature importance analysis validated these symptoms as the largest risk prediction contributors, further supporting their clinical feasibility.

4.8 Temporal Pattern Recognition

The system performed very well in tracking temporal evolution of symptoms:

- Progressive Symptom Detection: 82.7% accuracy
- Symptom Clustering: 73.4% multiple symptoms identified in 6-month windows
- Long-Term Risk Estimation: 12.3% better accuracy for predictions over the period of 12 or more months as compared to single-visit risk estimation

4.9 Clinical Utility Assessment

Five oncologists appraised system outputs via semi-structured interviews:

Assessment Criteria Mean Score:

Table 4. Clinician Assessment Scores (Scale 1-5)

Explanation Clarity	4.2
Clinical relevance	4.4
Trust-worthless	4.0
Decision Support Usefulness	4.3
Practice Change Potential	4.1
Overall Clinical Utility	4.2

Clinicians gave positive ratings across all domains, especially concerning clinical relevance and usefulness for decision support, indicating a potential benefit in real-world applications.

5 Discussion

It is also applied to detect early lung cancer in never-smokers, sacrificing patients of this type – the first breath of the second most common cancer in the world – by population-based screening guidelines, to which we are proposing a multi-agent AI paradigm in this paper. By integrating unstructured clinical narratives with symptom-gene associations and genomic features on a single knowledge graph driven platform, it offers an interpretable and clinical-motivated answer to risk prediction. The three-agent pipeline (symptom extraction, gene mapping and patient-gene scoring) was stable, reaching 85.7% accuracy, 88.3% sensitivity, and an ROC-AUC of 0.867, outperforming classical machine learning baselines by 10-20%. Additional clinical input was supportive of the interpretability and possible utility in the decision-support context. Although promising, this work is conducted on synthetic data, highlighting the need for clinical studies validation. Future studies will be to add multi-center data to prospectively validate our findings and to generalize the model for use in other populations. Further enhancements such as domain/specific LLM finetuning, dynamic knowledge graph updates, integration with imaging modalities, and uncertainty quantification will be investigated to improve model robustness. Further, the generalization of this platform to other forms of cancer, and potentially other chronic diseases, is sought to extend its clinical utility. In summary, this work opens the door for reliable, explainable agentic AI to preventive oncology.

5.1 System Architecture Overview

Methodology: We present a multi-agent architecture that combines temporal symptom modeling and knowledge graph representation in a joint explainable AI framework.

Technical: NLP for clinical notes, graph-based reasoning algorithms, Rule-based scoring with clinical interpretation, and automated end-to-end pipeline.

Clinical: Symptoms affected at least 50% of participants (fatigue 67%, dyspnea 64%, cough 58%), proved to effectively stratify risk, and were externally validated by clinicians with high scores.

5.2 System Architecture Overview

Clinical Validation: Multi-center cooperation for real clinical data validation, prospective studies, heterogeneous population inclusion, and long-term outcome follow-up.

Technical Improvements: Domain-specific fine-tuning, automated knowledge graph updating, medical imaging support, uncertainty quantification, and scalability enhancement.

Extended Applications: Customization for other malignancies, long-term disease surveillance, therapeutic response forecasting, and identification of side effects.

The success of this demonstration shows the promise of agentic AI for early cancer detection in an underserved group of never-smokers, and sets the stage for tackling general medical AI problems.

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